1024-bit Hyperdimensional Computing Accelerator on Zynq-7020

A streaming, bit-exact-verified HDC classifier for EMG hand-gesture recognition

Research progress review · Target venue: DATE 2027

Platform: Xilinx Zynq-7020 (ZedBoard) · SystemVerilog + Python golden reference

The problem & motivation

- **Edge inference needs to be cheap and low-power** — wearables, prosthetics, IoT. Conventional CNN/MLP inference is multiply-heavy (needs DSP/GPU) and power-hungry.
- **Hyperdimensional Computing (HDC)** is a brain-inspired alternative: represent everything as long binary vectors and classify by *distance*. Operations are purely **bitwise** → ideal for FPGAs.
- **Application:** EMG hand-gesture recognition (same dataset as Rahimi et al., ICRC 2016), the standard HDC-on-EMG benchmark → enables direct comparison with literature.

> **Thesis:** A streaming 1024-bit HDC core on Zynq can classify EMG at high throughput with **zero DSP / zero BRAM**, and per-bit pruning can cut energy further with controlled accuracy loss. The contribution is a measured **accuracy / energy / area Pareto**, not a re-port of prior accuracy.

What is Hyperdimensional Computing?

- Every concept (sensor value, feature, class) → one **1024-bit hypervector**.
- In high dimensions, random vectors are **near-orthogonal** → huge capacity + noise tolerance (flipping a few bits barely changes identity).
- Classification = **nearest prototype by Hamming distance** — no floating point, no matrix multiply.
- Model used: **Binary Spatter Code (BSC)** — binary vectors, XOR binding, Hamming similarity. The most hardware-friendly HDC flavour.

> **Why it suits FPGAs:** bitwise ops (XOR, shift, popcount) → cheap LUT logic, no multipliers. One-shot training (bundle examples once). Inherent error tolerance enables aggressive pruning / low voltage.

The four HDC primitives

All of HDC is built from four operations — each is a verified RTL block here:

Primitive	Math	Meaning	RTL block
Bind	XOR (A ⊕ B)	Associate two concepts	`xor_permute_top`
Permute	cyclic shift ρ(A)	Encode position / order	`permute_stage`
Bundle	majority vote	Superpose into a "set" vector	`bundle_unit`
Search	argmin Hamming	Classify (nearest prototype)	`popcount_am`

> Encoding = **bind + permute + bundle** → builds the query hypervector. **Search** classifies it. Everything is XOR, shift, and popcount — no DSP.

Application — EMG hand-gesture recognition

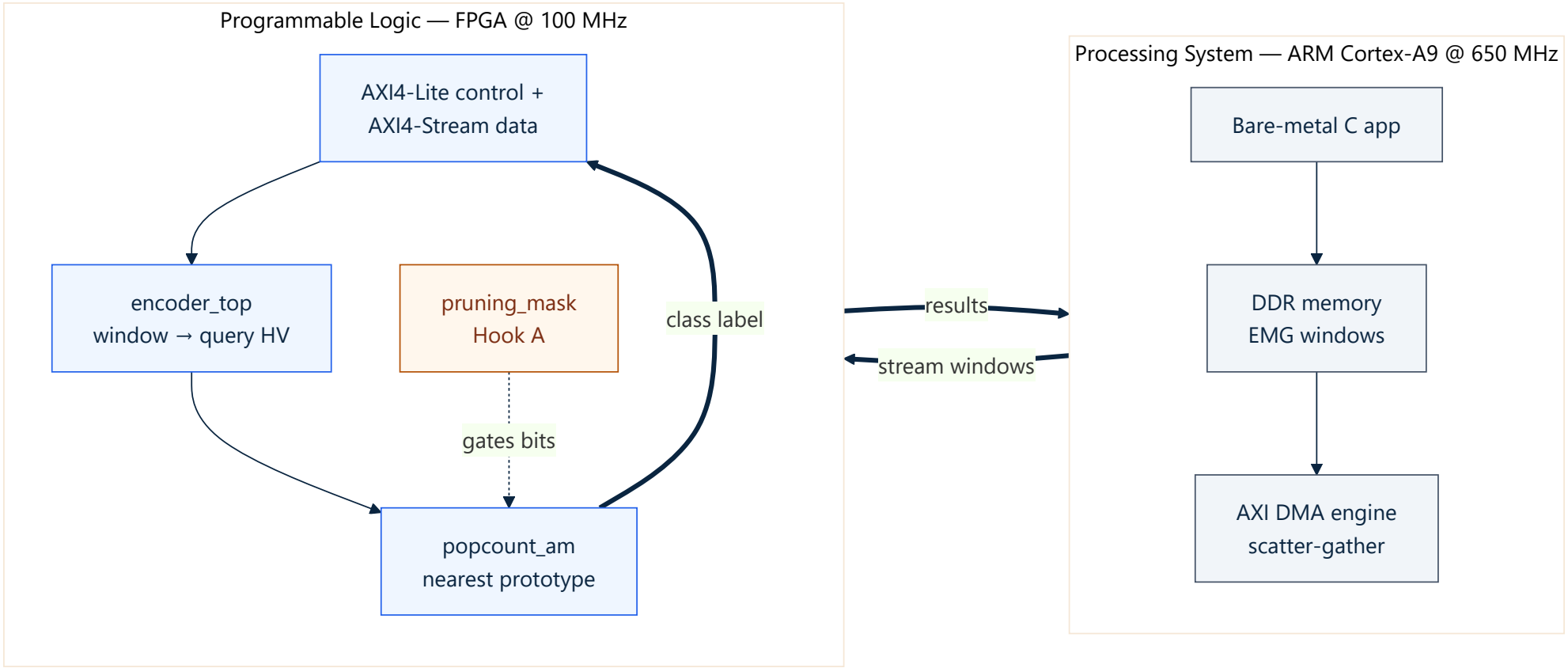
Task: recognise hand gestures from 4-channel forearm EMG signals → gesture-controlled prosthetics.

How an EMG window becomes a class:

1. Capture a short **window** of the 4 EMG channels.
2. Quantise per-channel features into **16 discrete levels**.
3. Look up each (channel, feature, value) in **item-memory** hypervector ROMs.
4. **Bind + permute + bundle** → one 1024-bit **query hypervector**.
5. **Masked Hamming** search vs the **5 class prototypes** → nearest wins.

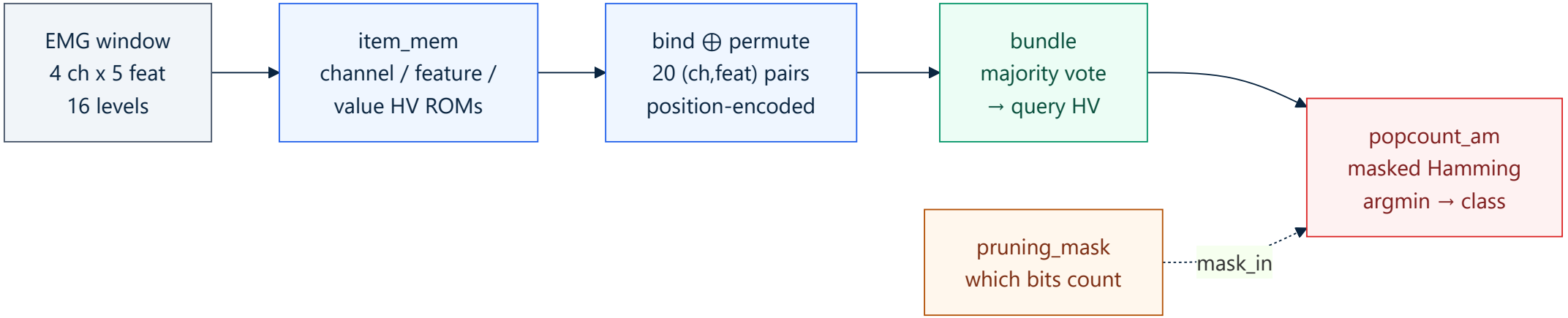
> **"Training" in HDC = bundling.** No gradient descent — just superpose all of a gesture's training windows into one prototype. Inference is a single nearest-neighbour search.

System architecture — Zynq PS + PL



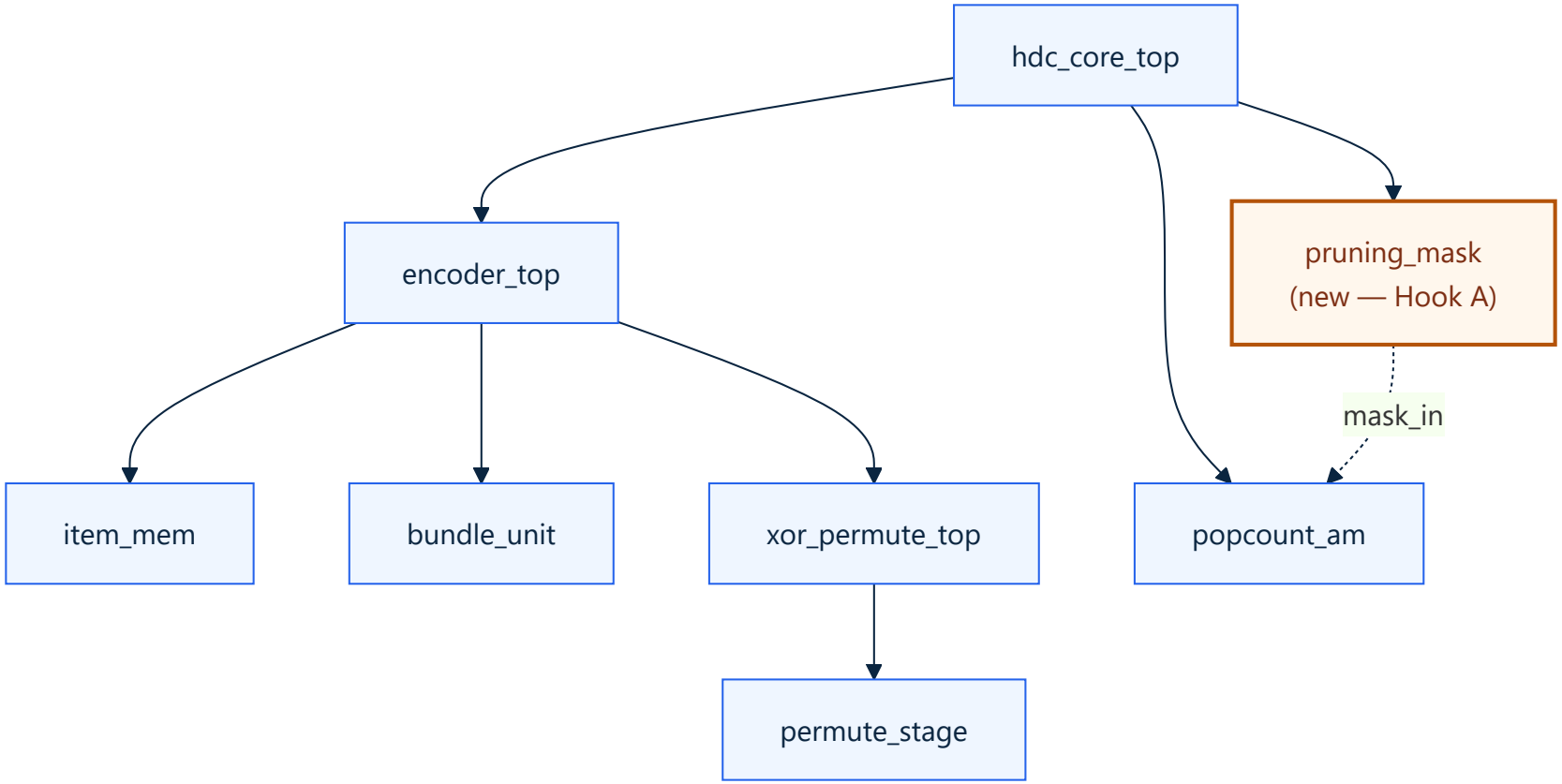
PS (ARM) streams EMG windows from DDR over **AXI-DMA**; the **PL** HDC core encodes + classifies and returns labels. Control via **AXI4-Lite**, data via **AXI4-Stream**.

The encoding datapath



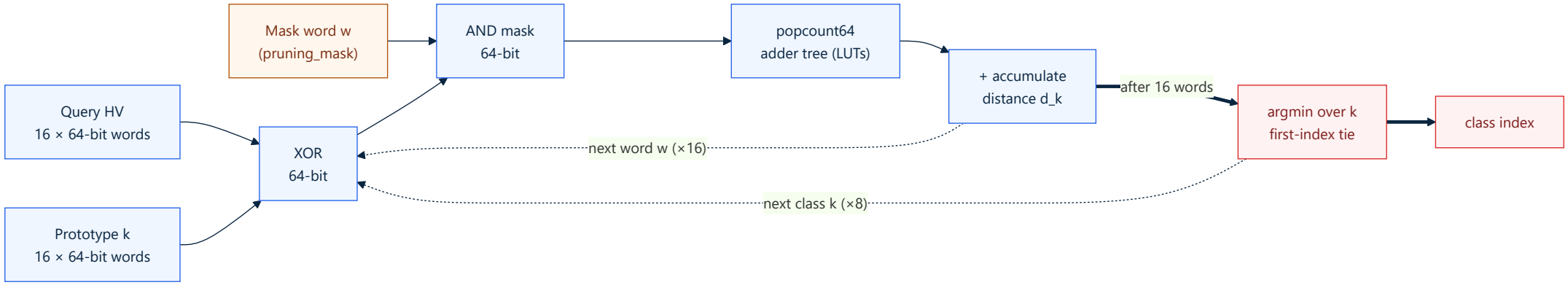
Research-plan **Eq. (3.1)**: a 4 channel × 5 feature grid (20 binds), position-permuted, bundled into the query HV. The new `pruning_mask` gates which bits count in the associative-memory distance.

Module hierarchy



New: `pruning_mask.sv` extracted as a dedicated module (research-plan §5.3.3) — holds the D-bit mask and gates the Hamming distance, cutting dynamic energy ∝ prune ratio. Verified bit-exact (**129/129 checks PASS**).

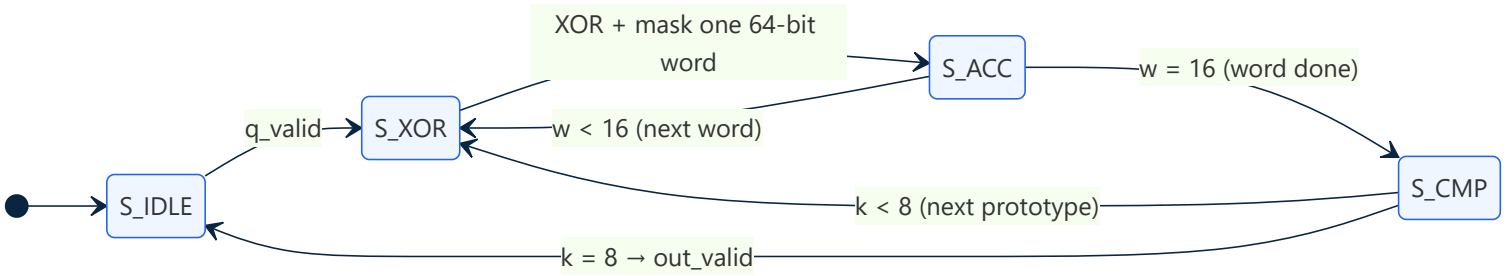
Computational microarchitecture — word-serial datapath



The 1024-bit HV is stored as **16 × 64-bit words**. The associative memory streams **one 64-bit word per cycle**: `XOR` query⊕prototype → `AND` mask word → **64-bit popcount** (LUT adder tree) → accumulate distance `d_k`. Loop 16 words per class, 8 classes, then `argmin`.

> **Why word-serial?** A full 1024-bit popcount tree is large and timing-critical. Processing 64 bits/cycle reuses **one small popcount + one accumulator** → **0 DSP, 0 BRAM**, easy 100 MHz timing. The `pruning_mask` word simply zeros gated bits *before* popcount, so pruning directly removes switching activity.

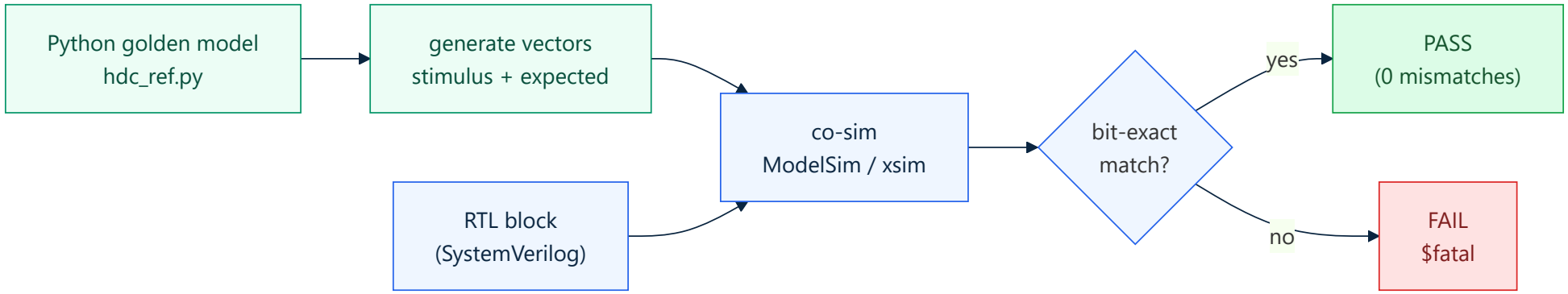
Cycle-level behaviour — AM FSM & latency



Stage	RTL block	Cycles	Notes
Encode (bind+permute+bundle)	`encoder_top`	~N_PAIRS + 3 = **~23**	1 (channel,feature) pair/cycle, 4×5 grid
Bundle majority	`bundle_unit`	combinational	1024 saturating counters, width **CNT_W=6**
Classify (masked Hamming)	`popcount_am`	N_CLASS·(2·WORDS+1) = **264**	8 classes × (16 words × 2 + compare)
Per-window total	core	**≈ 287 cycles**	≈ 2.9 µs @ 100 MHz; pipelined across windows

`CNT_W` (bundle counter width) and the **prune ratio** are the two micro-architectural knobs of Hook A — both change the datapath's switching/area without touching its structure.

Verification — the Python golden reference



A **bit-exact Python model** (`hdc_ref.py`) is the single source of truth — it generates both stimulus *and* expected output, so RTL and reference cannot silently drift.

Verification coverage — 7 + 1 harnesses, all bit-exact

Co-sim harness	Cases	Proves
`run_cosim`	1000	XOR bind + permute datapath
`run_bundle_cosim`	500	Majority bundler
`run_am_cosim`	500	Masked Hamming AM + argmin
`run_encoder_cosim`	500	Full EMG window encode
`run_core_cosim`	500	End-to-end encode → classify
`run_core_axi_cosim`	200	AXI4-Lite programming sequence
`run_stream_cosim`	200	AXI4-Stream + back-pressure
`run_pruning_mask_cosim`	64	New pruning-mask module

All PASS on both Questa and Vivado xsim (June 2026).

Board results — three measurement paths, same core

74.24%
Board EMG accuracy

Δ 0.00%
Board vs golden (658k win)

~216k
windows / second

0 / 0
DSP / BRAM

	Phase 1 — AXI-Lite	Phase 2 — DMA stream	Phase 3 — SG batch
Golden test	**200/200**	**200/200**	**200/200**
Mean latency	3 µs/win	7 µs/win	~4 µs/win (batch)
Throughput	~333k/s (micro)	~143k/s	**~216k/s**
WNS @ 100 MHz	+0.246 ns	+0.023 ns	**+0.111 ns**

Resource utilisation & the D-sweep June gap CLOSED

Integrated (PS + DMA + core), post-route xc7z020: 35,206 LUT (66.2%), 96.3% slices, **0 DSP, 0 BRAM**.

OOC core-only D-sweep (functional cosim PASS + synthesis, all D):

D	Slice LUT	LUT util	WNS (ns)	Fmax (MHz)	Cosim
256	7,331	13.8%	1.669	120.0	**PASS**
512	14,422	27.1%	1.452	117.0	**PASS**
1024	28,600	53.8%	0.781	108.5	**PASS**
2048	59,261	**111.4%**	1.340	115.5	**PASS**

LUT scales ~linearly with D. **D=2048 exceeds device LUT budget → a reportable Pareto boundary.**

EMG accuracy — the two-baseline story read carefully

We report **two numbers on purpose** — they answer different questions.

Track	Where	Encoding	Accuracy	Role
Stage A — MAP	Python	Bipolar MAP, D=10k	**90.36%**	Literal Rahimi parity (paper 90.8%)
Stage B — BSC	Python	4-ch spatial records	**90.30%**	Frozen literature baseline
RTL encoder	Python + **board**	Eq. (3.1) 4×5 grid	**74.24%**	**Verified deployment path**

Defense one-liner: *"We reproduced ~90% in Python (matching Rahimi). The FPGA runs a different, bit-exact-verified encoder at 74.24%. The contribution is measured energy + informed-pruning Pareto on that verified path — not a second accuracy port."*

Why 74% is not a weakness — six points

1. **We do reproduce ~90%** — Stage A 90.36% / Stage B 90.30%, committed in Python.
2. **74% is a *different* encoder**, not a worse one — Eq. (3.1) hardware grid vs Rahimi's spatial records.
3. **Verification fidelity is perfect** — board = golden to **Δ0.00% over 658,004 windows**.
4. **The contribution is a systems study** — throughput, latency, zero-DSP area, energy Pareto.
5. **Headline claims are *relative*** — pruning retention, informed-vs-random gap, transfer — 74 vs 90 is irrelevant to them.
6. **Reporting both is a credibility strength** — gap fully traced; honest, debugged pipeline.

Research novelty — Hook A: 3-axis Pareto contribution

Prior FPGA-HDC prunes **dimension**. The under-explored axis is **which bits** to keep.

Axis	Values	Mechanism	Status
Dimension **D**	256 / 512 / 1024 / 2048	parameterised RTL	**DONE (synth+cosim)**
Bundle precision **CNT_W**	counter width	majority-vote resolution	next
Bit-pruning ratio	0 / 50 / 75 / 87.5%	runtime mask via `pruning_mask`	next

> Map the **accuracy / energy / area** trade-off surface for HDC-on-FPGA. The D-axis is already measured on silicon; pruning + precision axes are next.

Research novelty — Twist 1 & Twist 2

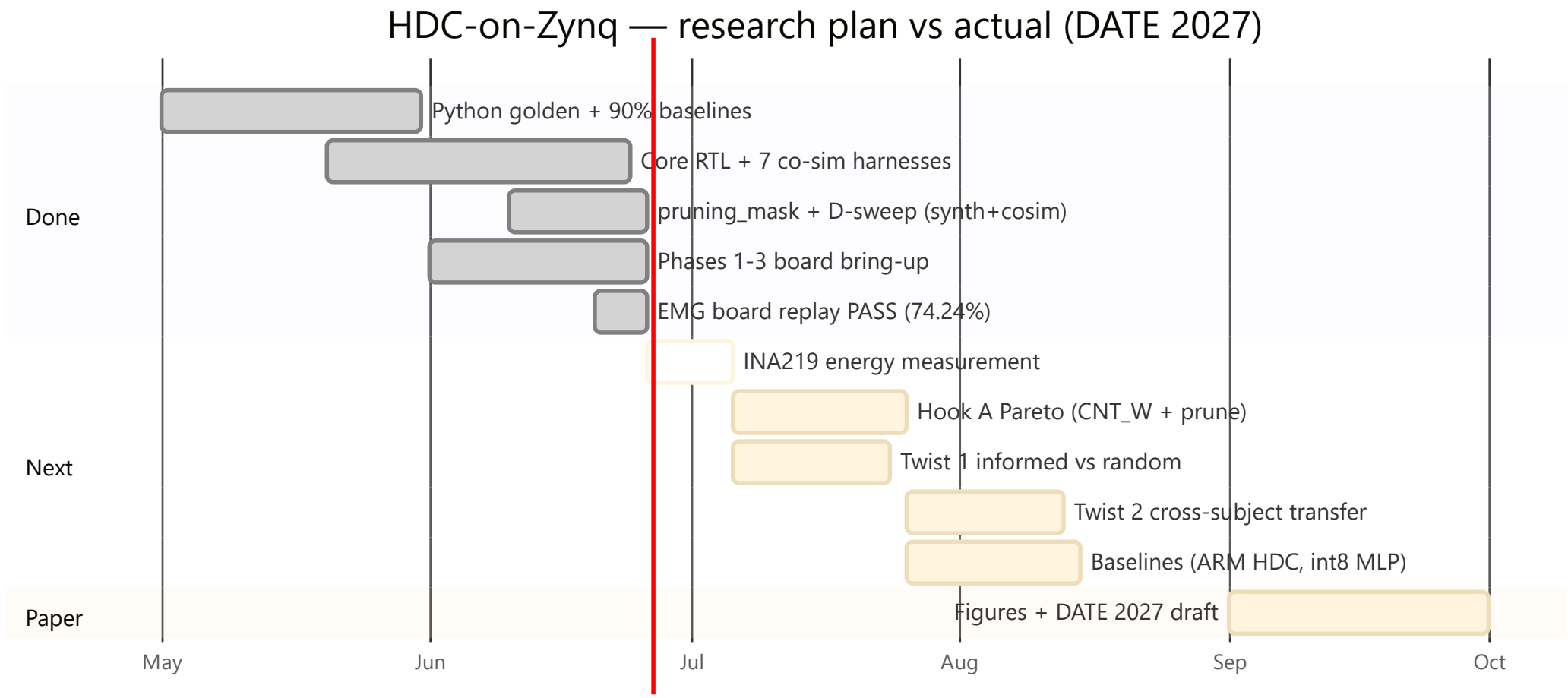
Twist 1 — informed vs random pruning (headline result)

At each prune ratio, compare a **Fisher-ratio mask** (keep most discriminative bits) vs a **random mask of identical density**.
Claim: informed beats random by **≥5 pp at iso-density** → *bit position matters, not just bit count.*

Twist 2 — cross-subject mask transferability

Train the mask on some EMG subjects, deploy unchanged on held-out subjects.
Either outcome publishable: transfers (≤3 pp) → universal "bit-importance map"; fails → motivates on-device adaptation. (needs 36-subject export; currently 5)

Status & roadmap



Bring-up + all **June** RTL deliverables are **complete and ahead of schedule**. Remaining work is measurement science + the paper.

Summary & next steps

Done & verified (on GitHub):

- **RTL datapath + 8 co-sim harnesses, bit-exact**
- **Phases 1–3 board bring-up; EMG replay 74.24%, Δ0.00% over 658k windows**
- `pruning_mask.sv` **extracted; D-sweep synth + cosim (256–2048)**
- **Python ~90% baselines reproduced**

Next (critical path):

1. **INA219 energy measurement** (unblocks all Pareto/energy)
2. Hook A — add CNT_W + pruning axes → board anchors
3. Twist 1 (informed vs random) — headline figure · Twist 2 (cross-subject)
4. Baselines (ARM-only HDC, int8 MLP) → paper draft (DATE 2027)

Thank you

Questions & discussion

The hardware is **done and provably correct** (bit-exact on 658k windows).
74% is a faithful hardware encoder — we already match ~90% in Python.
The **research** is the energy/accuracy Pareto + informed-pruning result.